Dipartimento di Ingegneria e Scienza dell'Informazione

– KnowDive Group –

KG 2025 - Project Report
Oncovirus Knowledge Graph

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2	November 11th 2025	Frederick Abi Chahine & Hala Alshaar	Purpose definition phase, including the informal purpose, scenarios, personas, CQs, concepts identification and ER model
3	November 22nd 2025	Frederick Abi Chahine & Hala Alshaar	Information gathering phase, including the data sources and data preprocessing
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1 Introduction

Oncoviruses, which are viruses that can induce cancer development, are responsible for a significant portion of human cancers. Examples include Human Papillomavirus (HPV), Hepatitis B and C viruses (HBV, HCV), and Epstein–Barr Virus (EBV). Understanding the molecular interactions between viral and host proteins, as well as their influence on cellular pathways and phenotypes, is essential to reveal oncogenic mechanisms and develop targeted therapies or preventive vaccines [1][4].

However, the knowledge surrounding oncoviruses is fragmented across multiple data sources (UniProt, KEGG, Reactome, literature). The relationships between viruses, proteins, pathways, phenotypes, and cancers are not easily integrated or queried in a unified way.

A Knowledge Graph (KG) provides a powerful framework to integrate this heterogeneous information into a connected structure that enables semantic reasoning, hypothesis generation, and advanced querying [2]. By linking data about viruses, host proteins, pathways, and cancer types, such a graph can reveal underlying biological mechanisms and support discoveries in precision oncology and preventive medicine. Thus, this project aims to construct a comprehensive oncovirus KG following the ITELOS methodology, which emphasizes clarity of purpose, precise domain definition, and structured knowledge design [3].

2 Purpose Definition

2.1 Informal Purpose

The purpose of this KG is to establish a unified and semantically rich representation of the biological mechanisms through which viruses contribute to cancer development. It seeks to integrate dispersed information about oncoviruses, their encoded proteins, and the host cellular components they interact with, in order to elucidate the molecular and systemic connections that underlie viral oncogenesis.

By connecting entities such as viruses, host and viral proteins, cellular pathways, phenotypes, cancers, and vaccines, the graph will enable researchers to navigate complex relationships that are otherwise scattered across literature and databases. The goal is to create a framework that not only supports data integration but also facilitates reasoning, discovery, and hypothesis generation.

This KG will serve as a bridge between molecular biology, computational modeling, and clinical understanding, allowing scientists to trace how a virus infects a host, manipulates its molecular machinery, disrupts cellular pathways, and ultimately drives tumorigenesis. Furthermore, it will provide a foundation for vaccine and therapeutic research by highlighting targets and pathways implicated in virus-induced cancers.

2.2 Domain of Interest

The domain of interest encompasses the molecular, cellular, and clinical dimensions of virus-induced cancers, focusing on the interactions between oncogenic viruses, host cellular machinery, cancer-related pathways, and therapeutics. The KG integrates biological, molecular, and clinical information to model how viruses contribute to tumorigenesis across different organisms and tissue types.

2.2.1 Space

The spatial scope of this study primarily involves human biological systems, where the focus lies on host–virus interactions at the molecular and cellular levels. Specifically, the graph captures protein interactions within cellular compartments and within tissues or organs that serve as the primary sites of infection and cancer development (e.g., liver for HBV/HCV, cervix for HPV, lymphatic tissue for EBV).

The viruses represented include major oncoviruses of clinical relevance, such as Human Papillomavirus (HPV), Hepatitis B Virus (HBV), Hepatitis C Virus (HCV), Epstein–Barr Virus (EBV), and Human T-lymphotropic Virus (HTLV-1), reflecting a global distribution across diverse populations and cancer types.

2.2.2 Time

The temporal scope extends from the initial viral infection to the development and progression of virus-induced cancers. This includes stages such as viral entry, genomic integration, latency, immune evasion, cellular transformation, and tumor formation.

While the KG is not bound to specific chronological datasets, it integrates findings from studies conducted between the years 2000 and 2025, encompassing both historical discoveries and contemporary research in virology, oncology, and bioinformatics.

2.3 Scenarios

These represent practical use cases demonstrating how the KG can be used.

1. **Identify Virus–Cancer Links:**
   A researcher queries which viruses are causally associated with specific cancer types (e.g., HPV → cervical cancer).

2. **Map Virus–Host Interactions:**
   A bioinformatician explores which human (host) proteins interact with viral proteins to identify potential therapeutic targets.

3. **Analyze Disrupted Pathways:**
   A molecular biologist traces which signaling pathways are affected by viral proteins to understand oncogenic mechanisms.

4. **Explore Vaccine Coverage:**
   A public health analyst checks which viruses have vaccines and which remain without preventive measures.

5. **Phenotype-based Discovery:**
   A virologist studies virus phenotypes and correlates them with cancer incidence or severity.

6. **Functional Context of Proteins:**
   A computational scientist investigates how host and viral protein functions/localizations relate

to pathway disruptions.

7. **Comparative Oncovirus Analysis:**
   An academic compares multiple oncoviruses to identify shared pathways, target proteins, or phenotypic traits.

2.4 Personas

Personas represent typical users of the KG and their goals.

1. **Dr. Elisa Marino – Virologist**
   Focus: Understanding viral protein functions and phenotypes.
   Uses KG to connect viral mutations with oncogenic potential.

2. **Dr. James Liu – Computational Biologist**
   Focus: Integrating datasets and analyzing protein–protein interaction networks.
   Uses KG to run queries linking virus-host interactions with pathway disruptions.

3. **Dr. Maria Rossi – Oncologist**
   Focus: Translating molecular data into clinical understanding.
   Uses KG to explore which viral infections are associated with specific cancers.

4. **Dr. Daniel Ortega – Vaccine Researcher**
   Focus: Designing or improving vaccines targeting oncoviruses.
   Uses KG to identify viral proteins that are ideal immunogenic candidates.

5. **Dr. Anna Becker – Molecular Pathologist**
   Focus: Examining cellular pathways impacted by viral infections in tumor samples.
   Uses KG to cross-reference altered pathways and cancer data.

6. **Prof. Samuel Green – Bioinformatics Educator**
   Focus: Teaching integrative omics and KG technologies.
   Uses KG as a pedagogical example of biological knowledge modeling.

7. **Dr. Rania Al-Hassan – Data Scientist**
   Focus: Developing machine learning models on biological graphs.
   Uses KG embeddings to predict novel virus–cancer associations.

2.5 Competency Questions (CQs)

These questions define what the KG must be able to answer.

1. **Virus–Cancer Relationships**
   Which viruses are known to cause specific types of cancer?

2. **Protein Interactions**
   Which host proteins interact with specific viral proteins, and what is their biological function?

3. **Pathway Involvement**
   Which pathways involve the host proteins that interact with viral proteins?

4. **Vaccine Coverage**
   Are there existing vaccines targeting a given oncovirus?

5. **Viral Phenotypes**
   What phenotypic traits (e.g., virulence, tropism) are associated with a specific virus?

6. **Virus Targeting and Cancer Association**
   What viruses target a specific host protein, and what cancers are they associated with?

7. **Pathway Disruption by Viral Proteins**
   Which viral proteins disrupt a specific host biological pathway?

2.6 Concepts Identification

2.6.1 Entity Types (ETypes) and Attributes

Entity Types	Categories	Description / Role in KG	Attributes
Virus	Core	Represents oncogenic viruses responsible for specific cancers.	virusCode, organismName, species, host
Cancer	Core	Represents cancers associated with viral infections.	cancerID, cancerType, primarySite
Vaccine	Core	Represents preventive vaccines targeting oncoviruses.	vaccineID, vaccineName, vaccineDescription, vaccineSource
ViralProtein	Common	Encoded proteins from oncoviruses that interact with host proteins.	viralUniprotID, viralGeneName, viralProteinName
HostProtein	Common	Human proteins targeted or affected by viral proteins.	hostUniprotID, hostGeneName, hostProteinName
Pathway	Common	Biological pathway or process involving host proteins.	pathwayKeggID, pathwayName, pathwayCategory, pathwayDescription, pathwaySource
ProteinAnnotation	Contextual	Stores metadata describing protein-specific properties (function, localization,	proteinAnnotationID, function, localization

		evidence).	
Phenotype	Contextual	Represents observable virus-related traits (e.g., virulence, tropism, infectivity).	phenotypeID, virusType, trait, phenotypeSource

Table 1*: Representation of each Entity Type with their categories, description and role, and their attributes in the KG.*

2.6.2 Relationships

- Vaccine **targets** → Virus
- Virus **causes** → Cancer
- Virus **hasPhenotype** → Phenotype
- Virus **encodes** → ViralProtein
- ViralProtein **interactsWith** → HostProtein
- HostProtein **hasAnnotation** → ProteinAnnotation
- HostProtein **involvedIn** → Pathway

2.7 ER diagram

Based on the defined CQs, we created the following Entity–Relationship (ER) diagram. We took into consideration all relevant biological and biomedical scenarios, prioritizing the oncovirology and cancer research areas of interest. To achieve this, we integrated data derived from public biomedical sources and viral oncology literature, focusing on the relationships between viruses, cancers, vaccines, and molecular interactions [1][4]. We carefully inspected each data source and corresponding schema, noting variables and features of potential interest. During the ER diagram development, we aimed to group and diversify features, ETypes, and properties in alignment with the formalized purpose of modeling virus–host–cancer interactions.

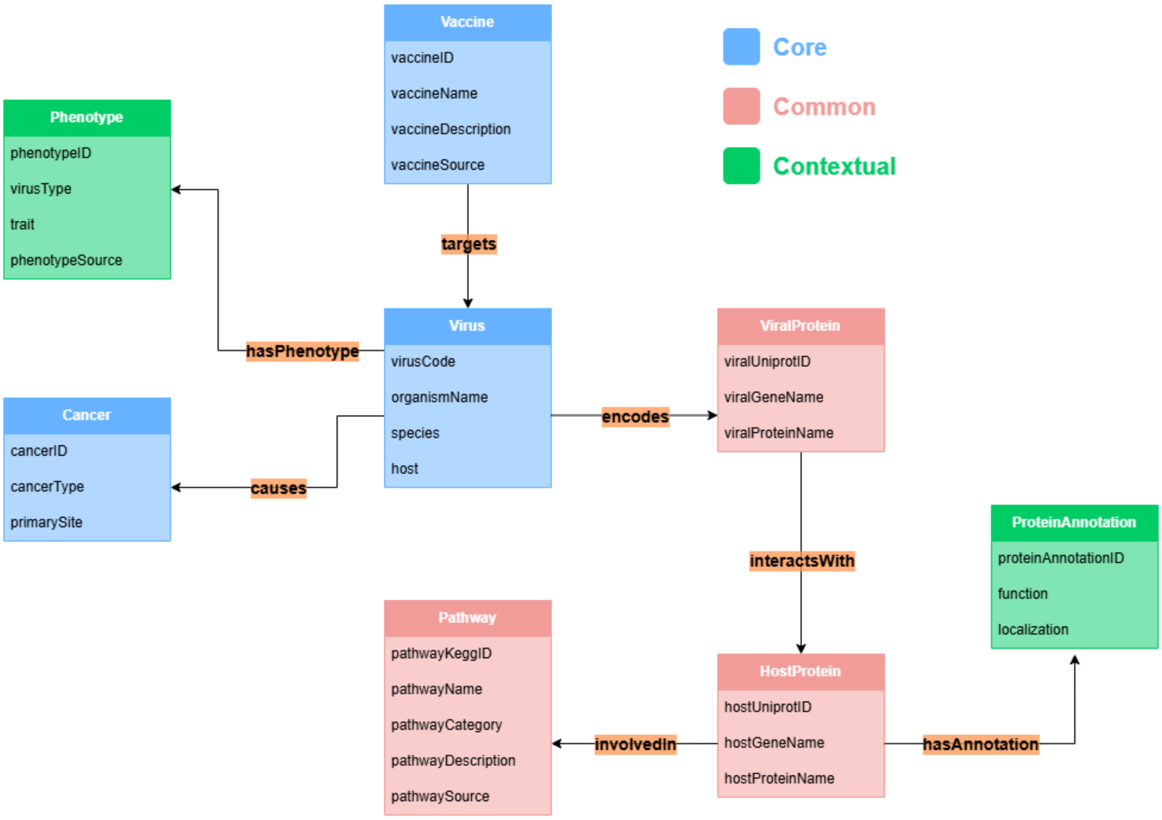


Figure 1: *ER diagram illustrating the ETypes with their attributes and relationships, where colors differentiate the EType categories: common, core and contextual, colored as red, blue and green respectively.*

2.7.1 EType Categories

We define the following ETypes and categorize them as Common, Core, or Contextual, following the same principles used in our domain analysis:

• Common ETypes:

1. **ViralProtein:** represents proteins encoded by oncoviruses that can directly or indirectly interact with host proteins. This EType is fundamental for capturing the molecular mechanisms through which viruses modulate host cellular pathways.
2. **HostProtein:** represents human proteins targeted or affected by viral proteins. This EType provides the necessary foundation to describe virus–host interactions at the molecular level.
3. **Pathway:** represents biological pathways or molecular processes involving host proteins. It enables grouping of host proteins into functional contexts, facilitating higher-level reasoning about disease mechanisms.

• **Core ETypes:**

1. **Virus:** represents oncogenic viruses responsible for specific cancers. This EType serves as a primary node in the KG, connecting to multiple other entities (cancers, vaccines, phenotypes and viral proteins).
2. **Cancer:** represents types of cancers associated with viral infections. It allows linking molecular mechanisms to clinical outcomes and supports CQs about oncogenesis and disease prevalence.
3. **Vaccine:** represents preventive therapeutics developed to target or neutralize specific oncoviruses. This EType helps address CQs related to prevention and immunization strategies.

• **Contextual ETypes:**

1. **ProteinAnnotation:** stores metadata describing protein-specific properties such as function, subcellular localization, and evidence type. This EType enriches the molecular host entities with additional biological meaning.
2. **Phenotype:** represents observable virus-related traits such as virulence, cell tropism, or infectivity. This EType is crucial for contextualizing viral behavior and linking it to pathogenic potential.

2.7.2 EType Relationships

We identify a set of biologically meaningful relationships capturing causation, interaction, and participation patterns within the system. Some are straightforward associative links (e.g., "Vaccine targets Virus"), while others represent complex molecular associations (e.g., "ViralProtein interactsWith HostProtein"). These edges are central to addressing the CQs concerning oncogenic mechanisms, molecular targets, and preventive interventions.

2.8 Qualitative Evaluation

During this initial phase, the informal purpose definition underwent a minor refinement following the construction of the preliminary ER model. This adjustment ensured a clearer alignment between the conceptual intent of the KG and the structural representation of the domain. The refinement helped articulate the objectives more precisely and confirmed that the emerging model accurately reflected the intended scope and analytical goals of the project.

3 Information Gathering

3.1 Data Sources

All the data necessary for this study are derived from multiple established bioinformatics and biomedical repositories [5][6][7][8][9][10]. These sources collectively provide comprehensive information on oncogenic viruses, including Hepatitis B virus (HBV), Human herpesvirus 8 (HHV-8), Epstein–Barr virus, B95 strain (EBV), Human papillomavirus strains 16 and 18 (HPV16, HPV18), and Human T-cell lymphotropic virus strains 1A and 1C (HTLV-1A, HTLV-1C), the viral proteins they produce, and their interactions with human host proteins. The following databases were integrated to construct the final dataset:

3.1.1 NCBI Virus Database

This dataset provides gene annotations, coding sequences (CDS), and protein products, serving as a foundation for linking viral genes to their protein functions and host targets [5].

3.1.2 UniProt Database

Supplies detailed protein-level information, including molecular functions, biological processes, and subcellular localization. Both viral and human host proteins are retrieved from UniProt to ensure consistent identifier mapping and functional annotation [6].

3.1.3 IntAct Molecular Interaction Database

Contains experimentally validated molecular interactions, particularly virus–host protein–protein interactions (PPIs) [7].

3.1.4 KEGG Database

Provides pathway and functional mapping for host proteins. KEGG data enable the contextualization of interactions within cellular pathways, facilitating interpretation of how viral infections disrupt host signaling and metabolic processes involved in cancer development [8].

3.1.5 ViralZone Database

This database provides biological context for each virus, helping to interpret how molecular interactions correspond to infection strategies and oncogenic potential [9].

3.1.6 World Health Organization (WHO)

Supply authoritative epidemiological data and official classifications of oncogenic viruses. WHO resources about recommended vaccines, allowing integration of public health information with molecular and interaction-level findings [10].

Each of these datasets contributes specific and complementary information:

- **Genomic and Protein Features (NCBI Virus, UniProt)**: gene, protein ids, and functional annotation.
- **Interaction Networks (IntAct)**: virus–host PPIs with evidence codes and literature references.
- **Pathway Integration (KEGG)**: mapping of viral effects on host cellular systems.
- **Biological Context (ViralZone)**: biological information about viruses
- **Epidemiological and Clinical Data (WHO)**: classification and global impact of oncogenic viruses, recommended vaccines

Together, these resources enable the construction of an integrated dataset encompassing viral–host molecular interactions, functional annotations, and cancer associations, forming the foundation for downstream analyses and predictive modeling of virus-induced carcinogenesis.

3.2 Data Preprocessing

During the data preprocessing phase, interaction information was collected for seven viral entities, derived from five major oncogenic viruses: HBV, HHV-8, EBV (B95 strain), and multiple strains of HPV (HPV16 and HPV18) and HTLV (HTLV-1A and HTLV-1C). Virus–host PPI data were retrieved from the IntAct Molecular Interaction Database. All extraction, merging, and compilation steps were automated using customized Python scripts to ensure efficiency, reproducibility, and comprehensive collection of interaction records. There was no misalignment in the data-gathering process. Following the generation of the fully merged IntAct dataset, an extensive data-cleaning procedure was applied to improve accuracy and ensure consistent annotation. In this step, 2,360 interaction records were removed due to missing required fields, and an additional 148 records were discarded as duplicates, defined as entries sharing the same combination of virus code, viral UniProt ID, and host UniProt ID. Only complete, unique, and biologically relevant interactions were retained for further analysis.

Following this cleanup, all remaining viral and host protein identifiers were cross-referenced with the UniProt database to retrieve detailed protein-level information, including functional annotations, sequence identifiers, and associated biological roles. Entries lacking sufficient UniProt information were excluded, resulting in the removal of 4 viral proteins and 82 host proteins. The final interaction dataset obtained after these filtering steps contained 31 distinct viral proteins and 759 human host proteins, representing a curated and well-validated set of interactions.

Subsequently, each host protein was mapped to its associated biological pathways using the KEGG and UniProt APIs in Python-based annotation workflows. The resulting pathway dataset underwent an additional round of cleaning to remove redundant, ambiguous, or incomplete pathway entries. Before this cleaning step, the pathway dataset consisted of 2,892 records; after filtering, 2,272 high-quality records remained, with 620 entries removed due to missing information or lack of clear annotation.

All final, cleaned, and biologically validated datasets have been saved in the repository as in the following table:

File	Content	Related EType
cancer_data.tsv	cancerID, cancerType, primarySite	Cancer
host_protein_data.tsv	hostUniprotID, hostGeneName, hostProteinName	HostProtein
viral_protein_data.tsv	virusCode, viralUniprotID, hostUniprotID, viralGeneName, viralProteinName	ViralProtein
protein_annotation_data.tsv	proteinAnnotationID, hostUniprotID, function, localization	ProteinAnnotation
vaccine_data.tsv	vaccineID, vaccineName, vaccineDescription, vaccineSource, targetVirus	Vaccine
virus_data.tsv	virusCode, organismName, species, host, causesCancer	Virus
virus_phenotype_data.tsv	phenotypeID, virusType, phenotypeSource, trait,	Phenotype

	virusCode	
pathway_data.tsv	pathwayKeggID, pathwayName, pathwayCategory, pathwayDescription, pathwaySource, hostUniprotID	Pathway

Table 2: *Datasets representing each EType, containing the final cleaned data.*

The GitHub repository contains several Python scripts used during the preprocessing and annotation of virus–host protein interactions:

1. **intact_datasets_merger.py**: Merging of virus–host PPI records from the IntAct Molecular Interaction Database; producing a single dataset.
2. **ppi_information_extractor.py**: Automates the extraction of PPI information from the fully merged intact dataset and excluding any duplicated records or ones with missing data.
3. **viral_and_host_protein_extractor.py**: Extracts and compiles the final set of viral and host proteins for downstream analysis, ensuring that only complete, unique, and biologically relevant proteins are included.
4. **protein_annotation_extractor.py**: Retrieves detailed protein-level information for host proteins by cross-referencing UniProt. It collects functional annotations, sequence identifiers, and associated biological roles.
5. **pathway_information_extractor.py**: Maps host proteins to their associated KEGG biological pathways via the KEGG and UniProt APIs.

3.3 Qualitative Evaluation

Throughout the information gathering phase, the ER model was revisited and refined to ensure coherence with the collected datasets. Several attributes that were no longer meaningful or relevant were removed, while others were renamed to improve clarity and consistency with biological terminology and data sources. Although certain relationship labels between ETypes were adjusted for precision, all previously defined relationships were retained, indicating that the original structural assumptions remained valid. Importantly, the informal purpose definition required no modification, demonstrating that the scope and objectives established in the purpose definition phase continued to align well with the data and domain understanding gained during this phase.

4 Schema and Ontologies

The cleaned datasets contain a range of entities and properties essential for analyzing oncoviruses and the interactions between viral and host proteins that disrupt cellular pathways leading to cancer. While most ETypes do not have suitable representations in *Schema.org* or *Bioschemas.org*, we were able to identify corresponding concepts for many of them in *NCBO BioPortal* under specialized biomedical ontologies [11][17][18]. When applicable, these ETypes are aligned with the established ontology terms; otherwise, we rely on schema derived directly from the original datasets.

1. Cancer *(from the Human Disease Ontology, DOID:162 via NCBO BioPortal)*

- **cancerID** (Integer, from the original data): Unique identifier for the cancer entry.
- **cancerType** (Text, from the original data): Category or type of cancer.

- **primarySite** (Text, from the original data): Anatomical site or primary location associated with the cancer type.

2. HostProtein *(from the original data)*

- **hostUniprotID** (Text, from the original data): Unique reference to the UniProt Accession Number of the host protein.
- **hostGeneName** (Text, from the original data): Gene name of the host protein.
- **hostProteinName** (Text, from the original data): Protein name of the host protein.

3. ViralProtein *(from the original data)*

- **viralUniprotID** (Text, from the original data): Unique reference to the UniProt Accession Number of the viral protein.
- **viralGeneName** (Text, from the original data): Gene name encoding the viral protein.
- **viralProteinName** (Text, from the original data): Protein name of the viral protein.

4. ProteinAnnotation *(from the original data)*

- **proteinAnnotationID** (Text, from the original data): Unique identifier for the protein annotation entry.
- **function** (Text, from the original data): Functional role of the protein.
- **localization** (Text, from the original data): Protein's subcellular localization.

5. Vaccine *(from Vaccine Ontology, VO_0000001 via NCBO BioPortal)*

- **vaccineID** (Integer, from the original data): Unique identifier for the vaccine entry.
- **vaccineName** (Text, from the schema.org): Name of the vaccine**.**
- **vaccineDescription** (Text, from schema.org): Description of the vaccine.
- **vaccineSource** (URL, from the original data): A reference to the origin of the vaccine information, typically a webpage, database entry, report, or publication from which the vaccine's entity data was obtained.

6. Virus *(from NCBITAXON, STY/T005 via NCBO BioPortal)*

- **virusCode** (Text, from the original data): Unique code for the virus.
- **organismName** (Text, from the original data): Scientific or common name of the virus.
- **species** (Text, from Invasion Biology Ontology, NCIT_C45293, *via NCBO BioPortal*): Viral species name.
- **host** (Text, from the original data): Primary host organism.

7. Phenotype *(from bioschemas.org)*

- **phenotypeID** (Integer, from the original data): Unique identifier for the phenotype entry.
- **virusType** (Text, from the original data): Category or type of the virus.
- **phenotypeSource** (URL, from the original data): A reference to the origin of the phenotype information, typically a webpage, database entry, report, or publication from which the virus's phenotype entity data was obtained.
- **trait** (Text, from the original data): Described phenotypic trait.

8. Pathway *(from Pathway Ontology, PW_0000001, via NCBO BioPortal)*

- **pathwayKeggID** (Text, from the original data): Unique reference to KEGG pathway database.
- **pathwayName** (Text, from the original data): Name of the biological pathway.
- **pathwayCategory** (Text, from the Schema.org): Pathway classification.
- **pathwayDescription** (Text, from Schema.org): Description of the pathway.
- **pathwaySource** (URL, from the original data): A reference to the origin of the information, typically a webpage, database entry, report, or publication from which the pathway's entity data was obtained.

This schema captures the structure of the datasets while integrating *Schema.org* and *Bioschemas.org* standards when suitable, offering a comprehensive framework for analyzing oncoviruses and the interactions between viral proteins and host proteins that disrupt cellular pathways leading to cancer.

5 Language Definition

The primary goal of the language definition phase is to formally specify the concepts and information that will be included in the final KG to fulfill the objectives of this project. To achieve this, we first identified all relevant concept structures, including ETypes, relationships, and object properties, that needed formal specification [3]. Each identified concept was then aligned with existing ontologies, first searching on the *Universal Knowledge Core (UKC)* to examine whether the concept was already defined [12]. If a concept was not present, it was subsequently checked against additional specialized resources, and if still not found, it was defined as a custom concept specific to our project. Due to the nature of this study, many of the elements are highly domain-specific, for example the EType HostProtein and its properties, and therefore required a complete formal definition within our framework. Since our datasets contain numerous concepts specific to molecular biology, virology, and immunology, we relied on specialized ontologies to retrieve definitions and standard identifiers, completing our Language Resource Table (Table 3). In particular, we used ontologies from NCBO BioPortal [11].

5.1 Concept Identification

We produced the Language Resource Table, in which each identified concept is listed and organized into three columns:

1. **ConceptID:** This takes one of three formats depending on where the concept was found: a UKC ID (ID: UKC-number) if present in UKC, a link to an external ontology if found elsewhere, or a custom project ID (ID: OKG25-#) if defined internally.
2. **Word-en:** The name or word, in English, associated with the concept.
3. **Gloss-en:** A clear explanation or definition for each concept in English.

To improve readability, we applied the following color-coding scheme:

- **Blue:** Represents ETypes from the ER Model
- **Green:** Represents Attributes from the ER Model
- **Red:** Represents Relationships from the ER Model

ConceptID	Word-en	Gloss-en
UKC-6543	Virus	Ultramicroscopic infectious agent that replicates itself only within cells of living hosts; many are pathogenic; a piece of nucleic acid (DNA or RNA) wrapped in a thin coat of protein.
UKC-24644	Vaccine	Immunogen consisting of a suspension of weakened or dead pathogenic cells injected in order to stimulate the production of antibodies.
UKC-26778	Phenotype	What an organism looks like as a consequence of the interaction of its genotype and the environment.
UKC-67961	Cancer	Any malignant growth or tumor caused by abnormal and uncontrolled cell division; it may spread to other parts of the body through the lymphatic system or the blood stream.
http://purl.obolibrary.org/obo/COMO_0000067	ProteinAnnotation	Additional information and metadata on proteins.
http://ncicb.nci.nih.gov/xml/owl/EVS/Thesaurus.owl#C17254	ViralProtein	Proteins encoded by viral genes.
http://purl.obolibrary.org/obo/IDOMAL_0001068	HostProtein	A protein that is produced by a host.
http://www.biopax.org/release/biopax-level3.owl#Pathway	Pathway	A set or series of interactions, often forming a network, which biologists have found useful to group together for organizational, historic, biophysical or other reasons. Usage: Pathways can be used for demarcating any subnetwork of a BioPAX model. It is also possible to define a pathway without specifying the interactions within the pathway. In this case, the pathway instance could consist simply of a name and could be treated as a 'black box'. Pathways can overlap, i.e. a single interaction might belong to multiple pathways. Pathways can also contain sub-pathways. Synonyms: network, module, cascade, Examples: glycolysis, valine biosynthesis, EGFR signaling.
UKC-695	localization	A determination of the place where something is.
UKC-6810	host	An animal or plant that nourishes and supports a parasite; it does not benefit and is often harmed by the association.
UKC-105757	function	Serve a purpose, role, or function.
http://edamontology.org/data	organismName	The name of an organism (or group of organisms).

_2909		
http://edamontology.org/data_1045	species	The name of a species (typically a taxonomic group) of organism.
http://ncicb.nci.nih.gov/xml/owl/EVS/Thesaurus.owl#C85496	trait	Any genetically determined characteristic.
http://ncicb.nci.nih.gov/xml/owl/EVS/Thesaurus.owl#C164895	pathwayName	Name of the biological pathway.
http://edamontology.org/data_2343	pathwayKeggID	Identifier of a pathway from the KEGG pathway database.
http://ncicb.nci.nih.gov/xml/owl/EVS/Thesaurus.owl#C164893	pathwayDescription	A brief explanation of what a pathway does or represents.
http://sbmi.uth.tmc.edu/ontology/ochv#C0449695	primarySite	Primary Site / Anatomical site or primary location associated with the cancer type.
http://edamontology.org/data_2592	cancerType	A type (represented as a string) of cancer.
http://purl.bioontology.org/ontology/SNOMEDCT/277234002	virusType	Describes the category or classification of a virus.
OKG25-01	virusCode	Unique identifier assigned to a virus.
OKG25-02	cancerID	Unique identifier for the cancer record.
OKG25-03	vaccineID	Unique identifier for the vaccine record.
OKG25-04	vaccineName	The official or common name of a vaccine.
OKG25-05	vaccineDescription	A brief explanation of what the vaccine does and what it targets or prevents.
OKG25-06	vaccineSource	The origin, database, or reference from which vaccine

		information is obtained.
OKG25-07	phenotypeID	Unique identifier for the virus phenotype record.
OKG25-08	phenotypeSource	The source or reference documenting the described phenotype.
OKG25-09	pathwayCategory	The broad functional class or grouping of a biological pathway.
OKG25-10	pathwaySource	The origin, database, or reference providing the pathway information.
OKG25-11	viralUniprotID	Unique Uniprot Reference to the viral protein.
OKG25-12	hostUniprotID	Unique Uniprot Reference to the host protein.
OKG25-13	viralGeneName	Standard name assigned to a viral gene.
OKG25-14	hostGeneName	Standard name assigned to a host gene.
OKG25-15	viralProteinName	Official or common name of a viral protein.
OKG25-16	hostProteinName	Official or common name of a host protein.
OKG25-17	proteinAnnotationID	Unique identifier for the protein annotation record.
UKC-98182	targets	Intend (something) to move towards a certain goal.
UKC-100755	causes	Give rise to; cause to happen or occur, not always intentionally.
OKG25-18	involvedIn	Indicates that an entity participates in, contributes to, or plays a role within a process, event, pathway, or higher-level activity. The relationship is generally directional, linking the participating entity (subject) to the broader process or context (object).
OKG25-19	interactsWith	Denotes that two entities engage in a direct or measurable interaction, influencing each other's state, function, or behavior. This relationship is typically symmetric, reflecting mutual participation in the interaction.
OKG25-20	hasAnnotation	Links an entity to a descriptive annotation or metadata.
OKG25-21	hasPhenotype	Links an entity to an observed trait, characteristic, or phenotype.
OKG25-22	encodes	Represents that a genetic element (such as a gene, transcript, or genomic region) of an organism specifies the sequence or structure of a functional product, typically a protein or RNA molecule. The relationship links the genetic source (subject) to the resulting biological product (object).

Table 3: *Language Resource Table where Concepts are listed with their IDs, English terms, and definitions. ConceptIDs come from the UKC, external ontologies, or project-specific OKG25 IDs. Colors indicate the ER model categories: ETypes, Attributes, and Relationships as blue, green and red respectively.*

5.2 Language Teleontology

In this step of the project, the focus was on constructing the language teleontology. Using the *Protégé* software [13], we formalized the language by defining the list of EType terms, object property terms, and data property terms as an OWL file. This included aligning our language to one or more reference language teleontologies, ensuring consistency and semantic clarity. Teleontology was then constructed to encode the domain language representation of our specific purpose, capturing object and data properties. The resulting OWL file, named oncovirus_KG_language_teleontology.owl, is available on our GitHub repository. This phase establishes a solid foundation by formalizing the structure and terminology of the language, preparing it for future integration into the complete KG and the subsequent formalization of the full teleology.

5.3 Qualitative Evaluation

During the language definition phase, one relationship in the ER diagram, linking Pathway to Cancer, was removed due to inconsistencies identified in the underlying data. This adjustment did not result in any loss of meaningful information, as the association can still be inferred indirectly through host proteins and their involvement in cancer-related pathways. The informal purpose of the KG remained fully aligned with the original intent, confirming the robustness of the conceptual design. In refining the schema, several relationships and attributes were renamed to improve semantic clarity, and redundant identifiers were removed where authoritative external IDs already served as unique identifiers. Corresponding adjustments were made in the datasets to ensure naming consistency and conciseness. The evaluation also revealed the need for a small number of supplementary attributes, which were retrieved and integrated without difficulty. Collectively, these refinements strengthened the coherence and precision of the schema while maintaining full compatibility with the biological evidence and data sources.

6 Knowledge Definition

The goal of the Knowledge Definition phase is to transform the informal semantic structures described in the Language Teleontology into a formalized, machine-interpretable Knowledge Teleontology. In this phase, the focus is on refining the conceptual model by assigning appropriate domains and ranges, formalizing ETypes, and defining object and data properties in accordance with the ITELOS methodology [3].

To achieve this, we systematically evolved the language-level model into a knowledge-level ontology by enriching each element with semantic constraints, explicit IRIs, and logical axioms. This ensures that the ontology is both syntactically correct and semantically coherent, enabling downstream reasoning and interoperability with external datasets.

6.1 EType Formalization

We began the formalization process by introducing a root EType, named "Thing", acting as the universal superclass for all ETypes in the ontology. This provides a consistent hierarchical structure and ensures compatibility with standard ontology engineering practices.

For each EType identified in the informal ER/EER model, we performed the following steps:

- Created a new subclass of Thing representing the domain entity. No additional subclass hierarchies were required for this project, so all ETypes remain direct descendants of Thing.
- Updated the IRI of each class to follow the required naming format: http://knowdive.disi.unitn.it/etype#<EType_name>
- Added annotations to each EType, including:
  - Label
  - Definition
  - Source of the information, referencing the Language Resource Table (Table 3)

This ensures that each EType is uniquely defined, traceable to its data origin, and aligned with the semantic constraints of the knowledge layer.

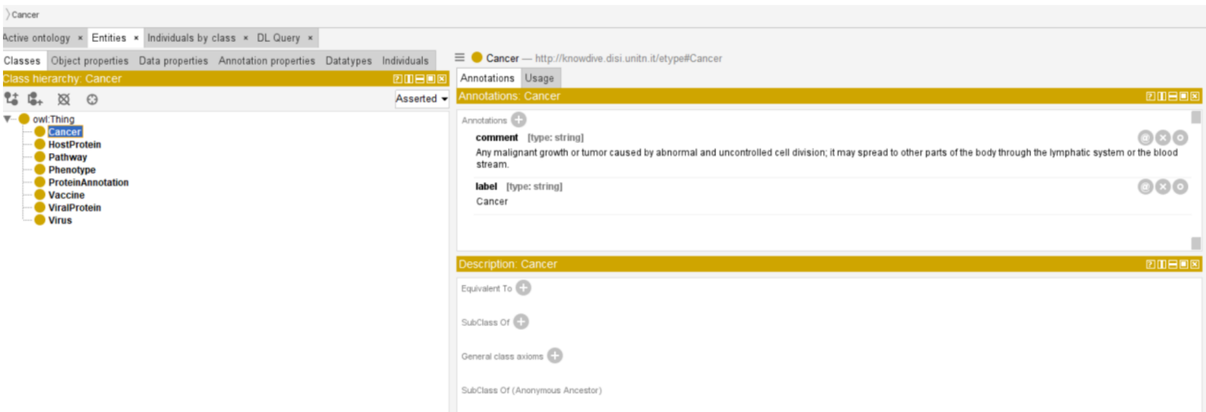


Figure 2: *Screenshot from Protégé displaying the formalization of ETypes. Each EType appears as a direct subclass of the root class Thing, following the required IRI pattern and annotated with its label, definition, and source reference. This view highlights the structured hierarchy of domain entities as represented in the ontology.*

6.2 Object Properties Formalization

Object properties formalize the relationships between ETypes and act as the structural backbone of the ontology. For each relationship defined in the language teleontology, we executed the following steps:

- Created a new object property named according to the language-level definition.
- Annotated the property with its definition and Concept ID.
- Updated its IRI using the required pattern: http://knowdive.disi.unitn.it/etype#has_<property-name>
- Specified the Domain(s) and Range(s) based on the ETypes involved in the relationship. This step formalizes directional semantics (i.e., indicating the class the relationship originates from and the class it points to).

This process enables the representation of inter-entity relationships in a structured and machine-interpretable form.

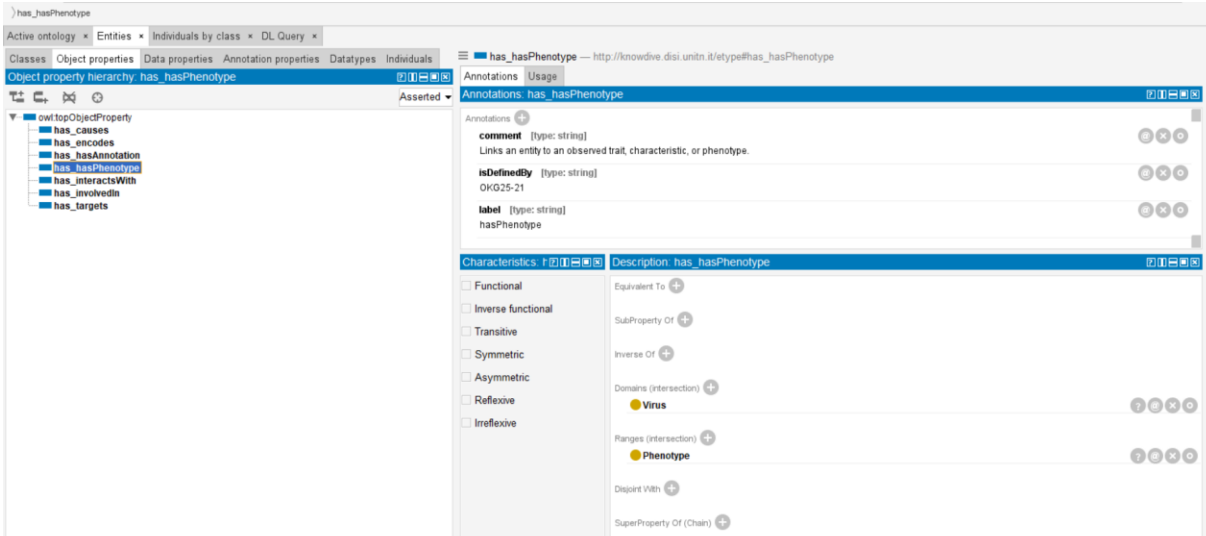


Figure 3: *Screenshot from Protégé showing the formalization of object properties. Each relationship from the language teleontology is represented as an object property with its assigned IRI, definition, and Concept ID. The Domain and Range fields specify the ETypes connected by the relationship, capturing the direction and semantics of the link.*

6.3 Data Properties Formalization

Data properties capture the literal attributes of each EType. For every data property extracted from the informal ER/EER model, we applied the following procedure:

- Created a new data property named according to the corresponding language-level label.
- Annotated it with its definition and Concept ID.
- Updated its IRI to follow the standardized format: http://knowdive.disi.unitn.it/etype#has_<property-name>
- Assigned Domain(s) corresponding to the EType that owns the attribute.
- Assigned Range(s) to reflect the datatype of the attribute (e.g., Boolean, String, Float).

This step ensures that each attribute is precisely defined and constrained, enabling validation and reasoning across instances.

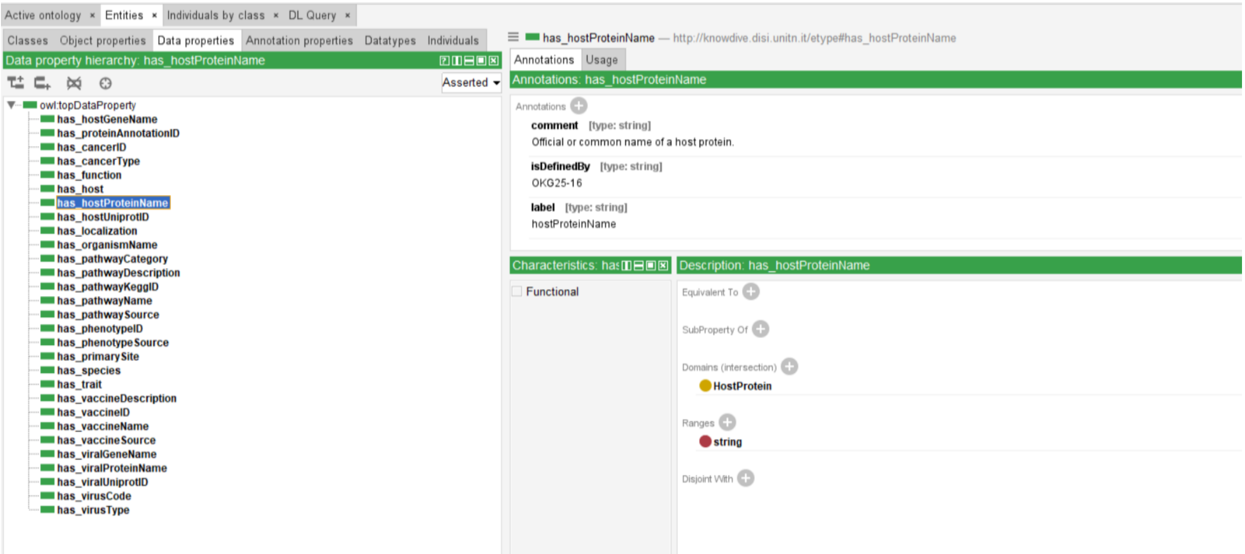


Figure 4: *Screenshot from Protégé showing the formalization of data properties. Each literal attribute from the ER/EER model is represented as a data property with its assigned IRI, definition, and Concept ID. The Domain specifies the EType that owns the attribute, while the Range indicates its datatype.*

6.4 Teleontology

Because this project focuses on a highly specialized biomedical domain, no existing teleontology fully matched our needs. Therefore, we constructed a custom teleontology using multiple ontologies available through the NCBO BioPortal [11].

Since we searched for exact term matches across BioPortal resources, we did not require ontology-to-ontology alignment procedures. However, the resulting teleontology still contains some redundancy; thus, instead of detailing each individual class and property, we provide a conceptual overview of the structure below.

6.4.1 Etype

Each EType was aligned with the most appropriate external ontology. Examples include:

- Cancer aligned using the Human Disease Ontology
- Virus aligned using the Semantic Types Ontology
- Vaccine aligned using the Vaccine Ontology

These alignments ensured consistency of terminology and compatibility with established biomedical knowledge sources.

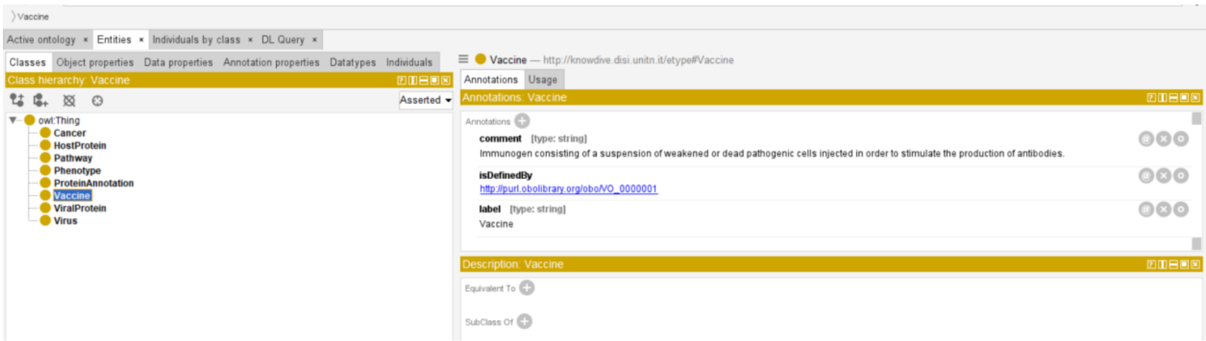


Figure 5: *Screenshot from Protégé showing the alignment of ETypes with external ontologies. Each EType is linked to the most appropriate reference ontology.*

6.4.2 Object Properties

For object properties, we were generally unable to locate exact matches within existing ontologies. As a result, most object properties were defined internally, based on the semantic requirements of the project.

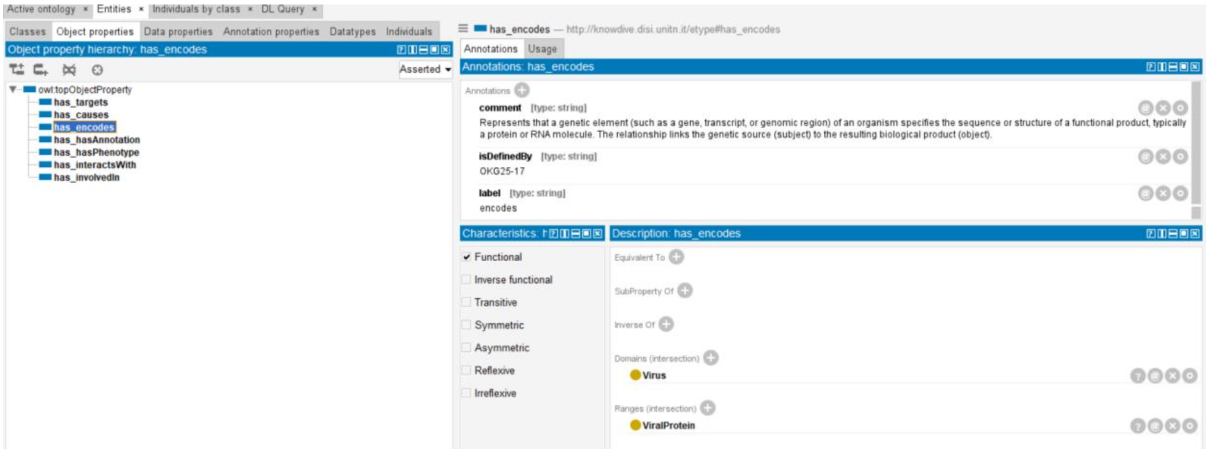


Figure 6: *Screenshot from Protégé showing object properties. Most properties were defined internally, as exact matches could not be found in existing ontologies.*

6.4.3 Data Properties

The external ontologies contained numerous attributes and branches that extended beyond the scope of our project. To maintain clarity and relevance:

- Only the relevant attributes were reused.
- A majority of data properties were defined manually, tailored specifically to our use case.

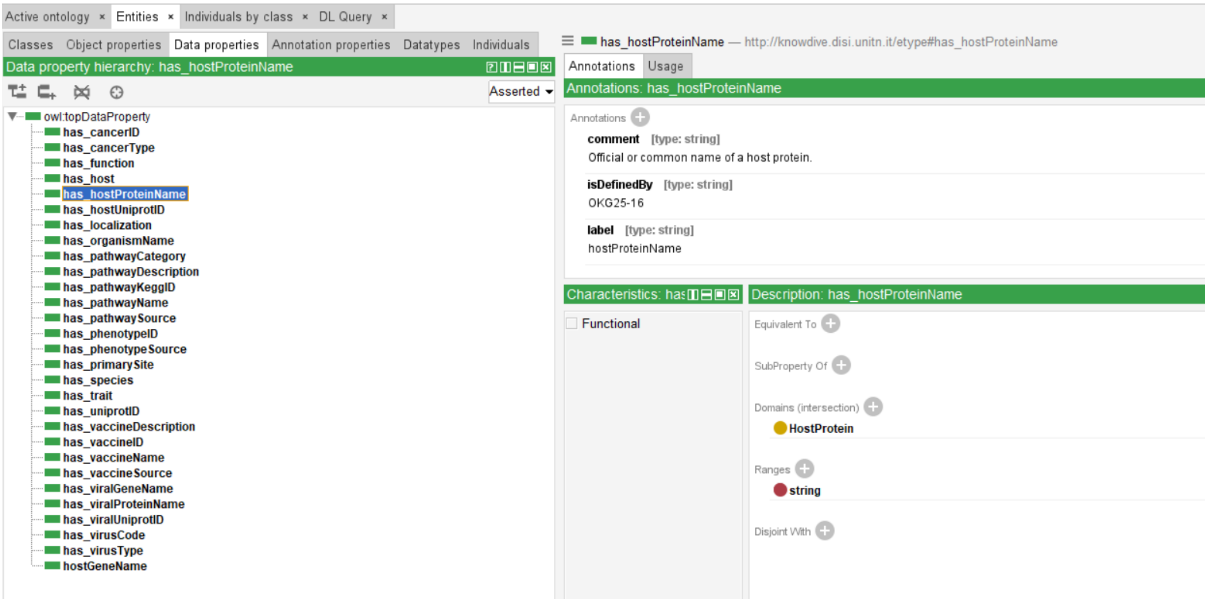


Figure 7: *Screenshot from Protégé showing data properties. While external ontologies contained many attributes beyond the project's scope, only relevant ones were reused.*

Our final teleontology (Figure 8) contains 8 ETypes (classes), along with 8 object properties and 29 data properties (counted as 8 and 30 due to an extra "intuition" header in these cases). The ontology also includes 4 annotation properties, 78 logical axioms, and a total of 261 axioms overall. The complete ontology was exported in OWL (.rdf) format and uploaded to GitHub under the name oncovirus_KG_knowledge_teleontology.owl.

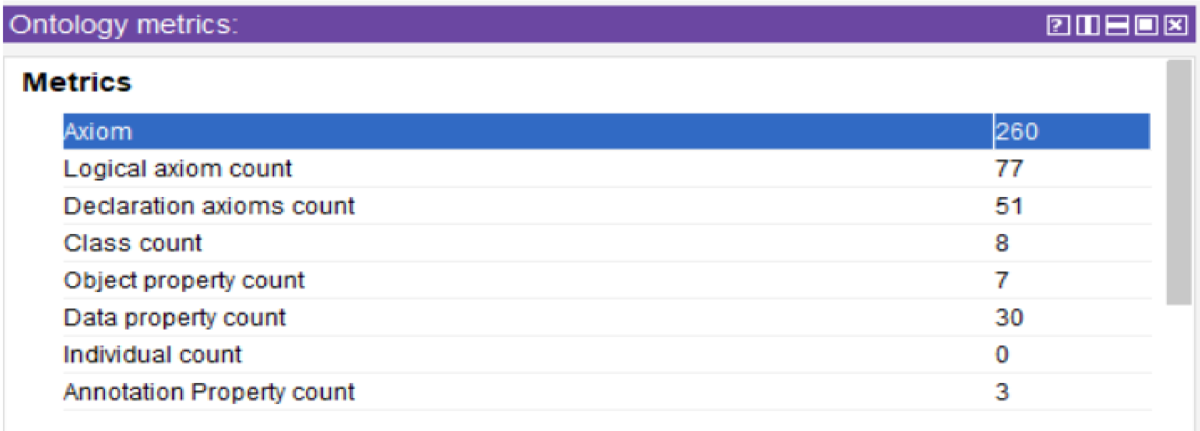


Figure 8: *Screenshot from Protégé showcasing the final teleontology.*

6.5 Qualitative Evaluation

During this phase, several refinements were introduced to improve consistency, integration, and compliance with the project guidelines. The virus–cancer lookup table was removed and replaced with an explicit attribute within the virus dataset that directly specifies the associated cancer type, resulting in a more streamlined representation of this relationship. To strengthen dataset interoperability, the viral protein dataset was enriched with host UniProt IDs, allowing interaction information to be incorporated directly rather than through pivot or lookup tables. Additionally, the IRIs within the language teleontology OWL file were revised to conform to the naming conventions required by the guideline sheet, ensuring semantic and structural uniformity. Finally, all object and data properties in the teleontology were updated so that their names begin with the prefix "has_" aligning them with the standard property-naming format prescribed by the ITELOS methodology.

7 Entity Definition

We are now entering a key phase of the project: defining and describing the entities we have identified, along with their attributes, relationships, and roles within the problem domain. The aim is to integrate the knowledge and data layers into a cohesive KG. At this stage, we work with clean and aligned datasets, and the previously developed teleology. To address remaining data heterogeneity, we follow a three-step process for entity matching, identification, and mapping, using *Karma* to accurately generate the final KG [14].

7.1 Entity Matching

Real world entities can appear differently across multiple datasets, requiring careful selection and consistent assignment of properties. However, thanks to our middle-out approach and the preprocessing performed in earlier phases, any potential misalignments have been resolved. As a result, entities consistently represent the same concepts, attribute values are coherent, and the overall structure aligns with the EER diagram. The datasets generated in previous steps are fully aligned, ensuring that all entities and their properties are accurately and consistently represented.

7.2 Entity Identification

The entity identification step focuses on selecting and defining the key entities that play a central role in our problem domain, while determining whether each entity has a unique identifier or an identifying set that can facilitate data integration and analysis.

For our KG, we identified the entities listed in Table 4. Each entity has an associated identifier that ensures unambiguous reference across datasets: for example, vaccineID for Vaccine, virusCode for Virus, and proteinAnnotationID for ProteinAnnotation. Some entities, such as ViralProtein and HostProtein, are closely related, but require separate identifiers (viralUniprotID and hostUniprotID) to distinguish their specific roles in the domain. Other entities, like Phenotype and Pathway, have dedicated identifiers (phenotypeID and pathwayKeggID) to allow precise tracking of associated biological or clinical information.

Entity	**Identifier**
Vaccine	vaccineID
Virus	virusCode
Phenotype	phenotypeID
ProteinAnnotation	proteinAnnotationID
ViralProtein	viralUniprotID

HostProtein	hostUniprotID
Pathway	pathwayKeggID
Cancer	cancerID

Table 4: *A table representing the main entities in the KG. Each entity has a unique identifier to ensure unambiguous referencing across datasets.*

In this section, our goal is to integrate the teleology with the actual data values from the datasets by mapping entities to their corresponding data representations and roles within the system. This involves capturing the interactions between entities through the previously defined relationships, while linking them to the appropriate columns and properties in the datasets. Another key objective is to present this information in a model graph using *Karma*, which facilitates a clear understanding of the structure and can be reused in future studies or research [14].

Karma is an integration tool designed to efficiently combine data from multiple sources and model it according to a chosen teleology (as defined in section 6). It also provides visualization of entity relationships across datasets and tracks modifications to enhance the reproducibility and usability of the work [14].

The output from *Karma* includes 8 main RDF Turtle (.ttl) files along with 8 R2RML models also in Turtle format for each of the 8 datasets described in Section 3, providing a comprehensive semantic representation of the data.

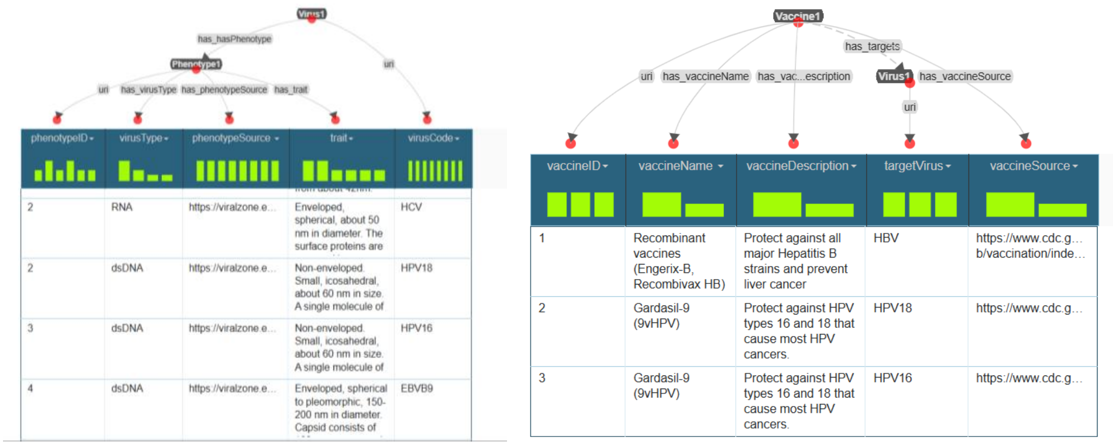


Figure 9: *Screenshots showing the integration of the teleontology with the datasets using Karma. Entities are mapped to their corresponding data representations and linked through the defined relationships.*

7.3 Visualization

After generating the RDF Turtle files using *Karma*, we imported them into *GraphDB* to support the visualization and inspection of different aspects of the KG [17]. This step enabled us to validate the structural correctness of the modeled entities and relationships, as well as to gain an overall understanding of the graph organization.

7.3.1 Class Relationship Visualization

We first examined the relationships between the different classes in the KG to verify that the modeled connections accurately reflect the intended ontology structure and semantic constraints (Figure 10).

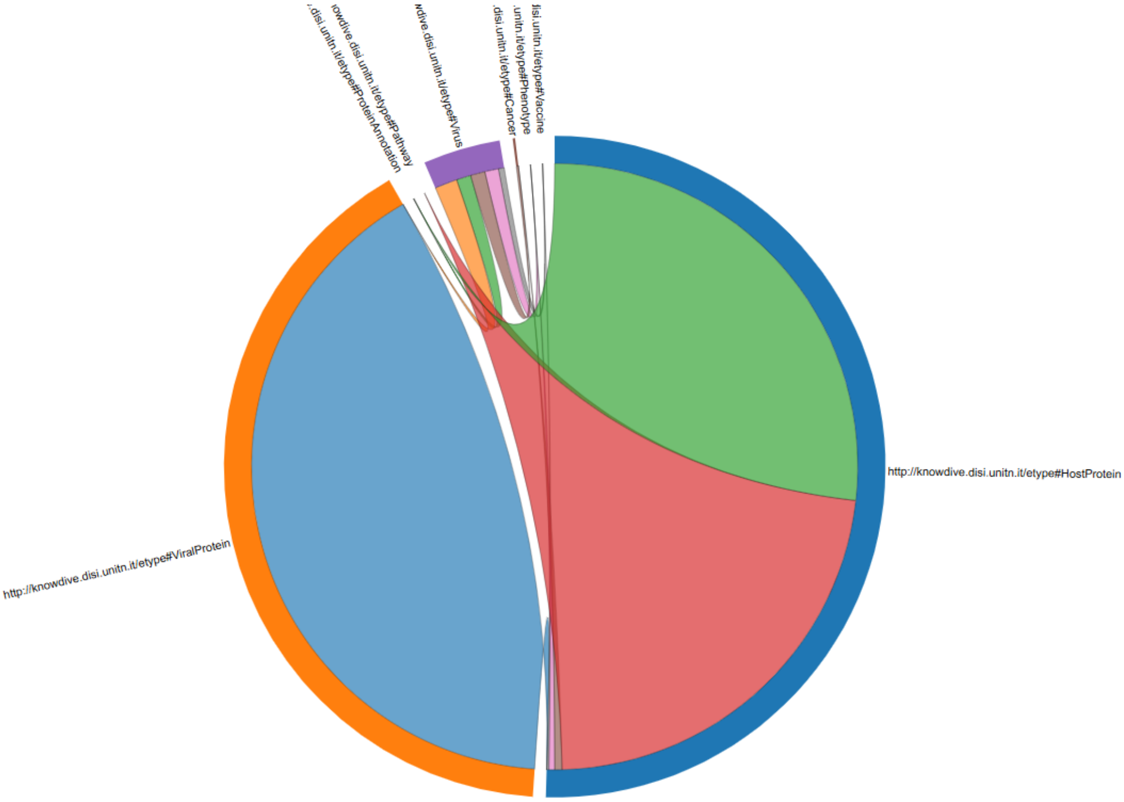


Figure 10: *GraphDB representation of the relationships between different entities.*

7.3.2 Graph Visualization

In addition, we generated and visualized a highly simplified overview of the KG to facilitate the interpretation of its overall structure and to better understand the general flow of information across entities and relationships (Figure 11).

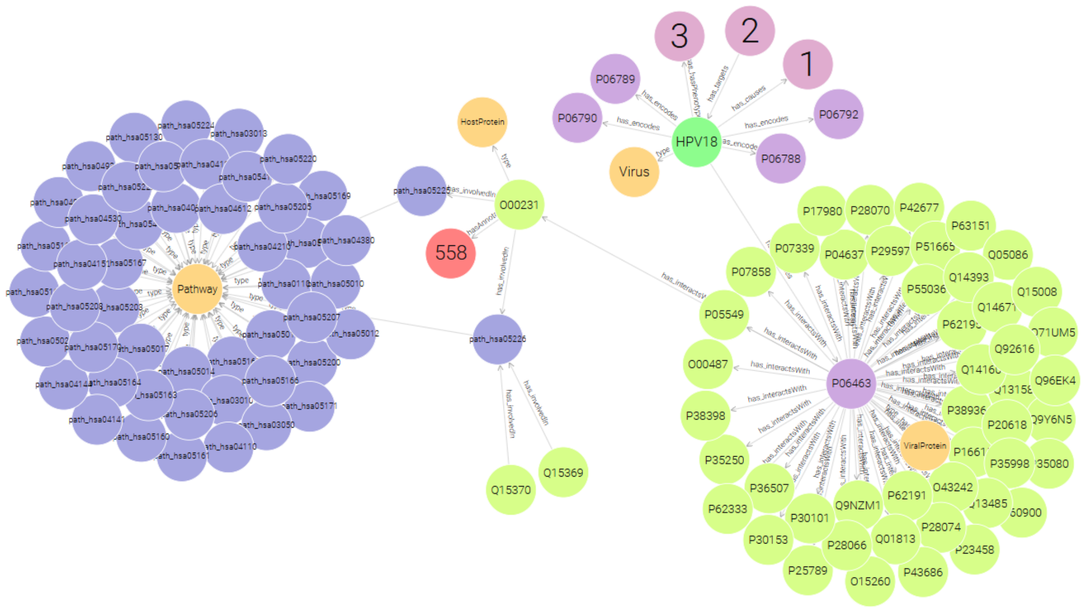


Figure 11: *Simplified GraphDB network representation of the KG, where each node contains the identifier of that instance.*

7.4 Qualitative Evaluation

During the visualization of the KG, we identified an issue with the hasAnnotation relationship between the ETypes HostProtein and ProteinAnnotation, as the relationship was not correctly instantiated in the graph. To resolve this problem, the mapping was revised and redefined within *Karma*, ensuring that the relationship was properly generated and represented in the resulting RDF output. In addition, we observed inconsistencies in the Pathway RDF Turtle file generated by *Karma*. Specifically, the KEGG pathway identifiers contained the prefix "path:", which caused parsing and interpretation issues within the tool. This problem was resolved by renaming the prefix to "path_", thereby restoring compatibility and allowing the pathway entities to be correctly processed and visualized.

8 Evaluation

Within the Knowledge Layer, two complementary evaluation objectives are defined. The primary objective focuses on assessing the alignment between the Teleontology and the CQs introduced in the initial project phases, with the goal of determining the extent to which the KG represents the relevant entities and relationships implied by those questions. The secondary objective extends this analysis by examining the correspondence between the teleontology and selected reference ontologies, enabling a broader comparison of conceptual coverage and semantic consistency.

8.1 Addressing CQs

To evaluate the performance of the KG, we performed 7 *SPARQL* queries in *GraphDB* to accurately address the CQs previously defined in section 2.5 [16]. Below are the figures representing each CQ along with its respective query and output result.

CQ1 - Which viruses are known to cause specific types of cancer?

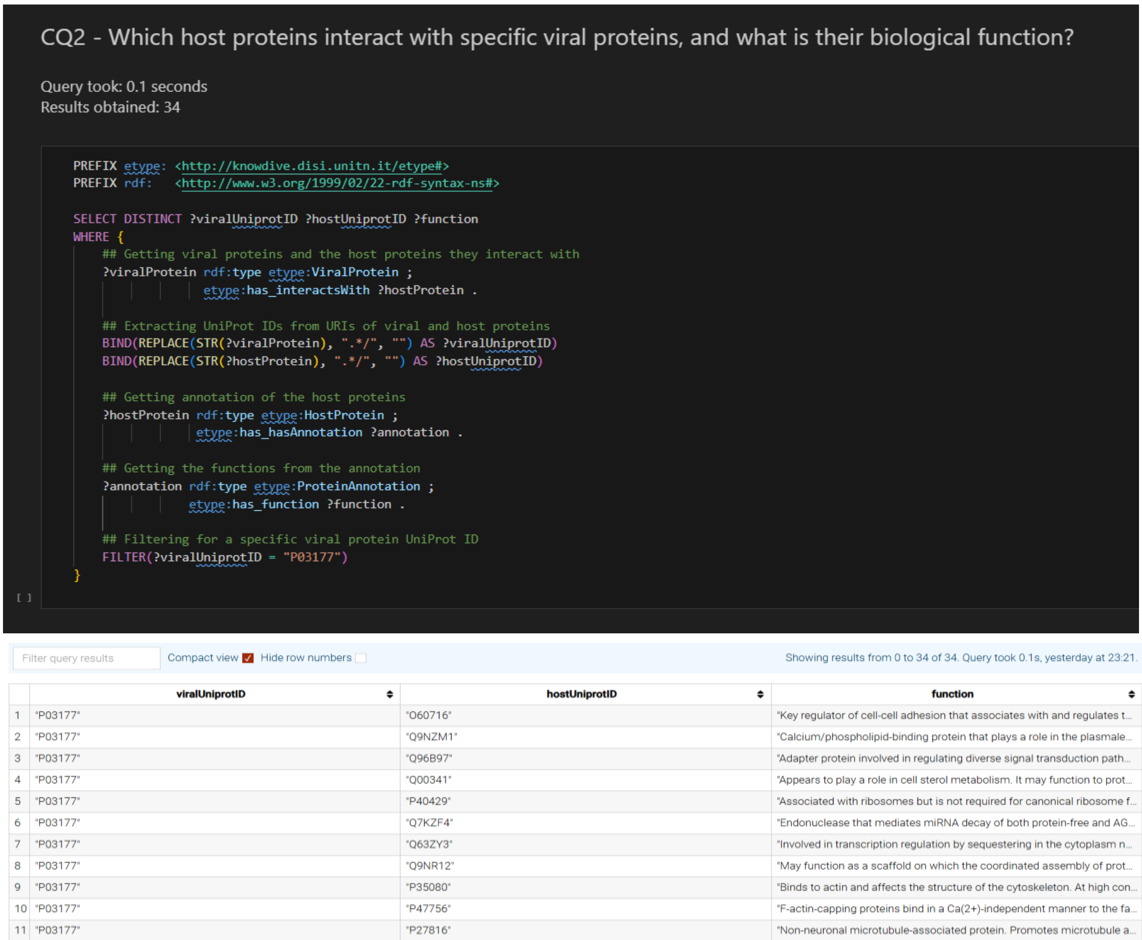
Query took: 0.1 seconds
Results obtained: 3

```
PREFIX etype: <http://knowdive.disi.unitn.it/etype#>
PREFIX rdf:   <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT DISTINCT ?virusName ?virusCode ?cancerType
WHERE {
    ## Getting the cancer that each virus causes
    ?virus rdf:type etype:Virus ;
            etype:has_causes ?cancer ;
            etype:has_organismName ?virusName .

    ## Extracting virusCode from its URI
    BIND(REPLACE(STR(?virus), ".*/", "") AS ?virusCode)

    ## Getting cancer type information
    ?cancer rdf:type etype:Cancer ;
             etype:has_cancerType ?cancerType .

    ## Filtering for specific cancer types
    FILTER(LCASE(STR(?cancerType)) = LCASE("Squamous cell carcinoma") || LCASE(STR(?cancerType)) = LCASE("Burkitt lymphoma"))
}
```

Filter query results | Compact view | Hide row numbers | Showing results from 0 to 3 of 3. Query took 0.2s, moments ago.

	virusName	virusCode	cancerType
1	"Human papillomavirus"@en	"HPV18"	"Squamous cell carcinoma"@en
2	"Human papillomavirus"@en	"HPV16"	"Squamous cell carcinoma"@en
3	"Epstein-Barr virus "@en	"EBVB9"	"Burkitt lymphoma"@en

Figure 12: *SPARQL query addressing CQ 1 along with the results of the query.*



CQ2 - Which host proteins interact with specific viral proteins, and what is their biological function?

Query took: 0.1 seconds
Results obtained: 34

```
PREFIX etype: <http://knowdive.disi.unitn.it/etype#>
PREFIX rdf:   <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT DISTINCT ?viralUniprotID ?hostUniprotID ?function
WHERE {
    ## Getting viral proteins and the host proteins they interact with
    ?viralProtein rdf:type etype:ViralProtein ;
                  etype:has_interactsWith ?hostProtein .

    ## Extracting UniProt IDs from URIs of viral and host proteins
    BIND(REPLACE(STR(?viralProtein), ".*/", "") AS ?viralUniprotID)
    BIND(REPLACE(STR(?hostProtein), ".*/", "") AS ?hostUniprotID)

    ## Getting annotation of the host proteins
    ?hostProtein rdf:type etype:HostProtein ;
                 etype:has_hasAnnotation ?annotation .

    ## Getting the functions from the annotation
    ?annotation rdf:type etype:ProteinAnnotation ;
                etype:has_function ?function .

    ## Filtering for a specific viral protein UniProt ID
    FILTER(?viralUniprotID = "P03177")
}
```

Filter query results   Compact view   Hide row numbers   Showing results from 0 to 34 of 34. Query took 0.1s, yesterday at 23:21.

	viralUniprotID	hostUniprotID	function
1	"P03177"	"O60716"	"Key regulator of cell-cell adhesion that associates with and regulates t...
2	"P03177"	"Q9NZM1"	"Calcium/phospholipid-binding protein that plays a role in the plasmale...
3	"P03177"	"Q96B97"	"Adapter protein involved in regulating diverse signal transduction path...
4	"P03177"	"Q00341"	"Appears to play a role in cell sterol metabolism. It may function to prot...
5	"P03177"	"P40429"	"Associated with ribosomes but is not required for canonical ribosome f...
6	"P03177"	"Q7KZF4"	"Endonuclease that mediates miRNA decay of both protein-free and AG...
7	"P03177"	"Q63ZY3"	"Involved in transcription regulation by sequestering in the cytoplasm n...
8	"P03177"	"Q9NR12"	"May function as a scaffold on which the coordinated assembly of prot...
9	"P03177"	"P35080"	"Binds to actin and affects the structure of the cytoskeleton. At high con...
10	"P03177"	"P47756"	"F-actin-capping proteins bind in a Ca(2+)-independent manner to the fa...
11	"P03177"	"P27816"	"Non-neuronal microtubule-associated protein. Promotes microtubule a...

Figure 13: *SPARQL query addressing CQ 2 along with the results of the query.*

CQ3 - Which pathways involve the host proteins that interact with viral proteins?

Query took: 0.1 seconds
Results obtained: 593

```
PREFIX etype: <http://knowdive.disi.unitn.it/etype#>
PREFIX rdf:   <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT DISTINCT ?hostUniprotID ?hostProteinName ?pathwayKEGG
WHERE {
    ## Getting viral proteins and the host proteins they interact with
    ?viralProtein rdf:type etype:ViralProtein ;
                  etype:has_interactsWith ?hostProtein .

    ## Getting pathways involving the host proteins
    ?hostProtein rdf:type etype:HostProtein ;
                 etype:has_hostProteinName ?hostProteinName ;
                 etype:has_involvedIn ?pathway .

    ## Extracting UniProt ID and KEGG pathway ID from URIs
    BIND(REPLACE(STR(?hostProtein), ".*/", "") AS ?hostUniprotID)
    BIND(REPLACE(STR(?pathway), ".*/", "") AS ?pathwayKEGG)
}
```

Filter query results Compact view Hide row numbers Showing results from 0 to 593 of 593. Query took 0.1s, moments ago.

	hostUniprotID	hostProteinName	pathwayKEGG
1	"Q06330"	"Recombining binding protein suppressor of hairless"	"path_hsa00010"
2	"O60812"	"Heterogeneous nuclear ribonucleoprotein C-like 1"	"path_hsa00030"
3	"O95816"	"BAG family molecular chaperone regulator 2"	"path_hsa00051"
4	"Q8TED1"	"Probable glutathione peroxidase 8"	"path_hsa00240"
5	"Q8TED1"	"Probable glutathione peroxidase 8"	"path_hsa00250"
6	"P35080"	"Profilin-2"	"path_hsa00330"
7	"P35080"	"Profilin-2"	"path_hsa00350"
8	"P35080"	"Profilin-2"	"path_hsa00380"
9	"P35080"	"Profilin-2"	"path_hsa00430"
10	"O75688"	"Protein phosphatase 1B"	"path_hsa00440"
11	"P12236"	"ADP/ATP translocase 3"	"path_hsa00480"
12	"P12236"	"ADP/ATP translocase 3"	"path_hsa00510"
13	"P12236"	"ADP/ATP translocase 3"	"path_hsa00512"
14	"P12236"	"ADP/ATP translocase 3"	"path_hsa00513"
15	"P47756"	"F-actin-capping protein subunit beta"	"path_hsa00562"
16	"O00483"	"Cytochrome c oxidase subunit NDUFA4"	"path_hsa00563"
17	"O00483"	"Cytochrome c oxidase subunit NDUFA4"	"path_hsa00564"
18	"O00483"	"Cytochrome c oxidase subunit NDUFA4"	"path_hsa00565"
19	"Q9Y5M8"	"Signal recognition particle receptor subunit beta"	"path_hsa00565"

Figure 14: *SPARQL query addressing CQ 3 along with the results of the query.*

CQ4 - Are there existing vaccines targeting a given oncovirus?

Query took: 0.1 seconds
Results obtained: 2

```
PREFIX etype: <http://knowdive.disi.unitn.it/etype#>
PREFIX rdf:   <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT DISTINCT ?virusCode ?virusName ?vaccineName ?vaccineDescription
WHERE {
    ## Getting vaccines, their information and the viruses they target
    ?vaccine rdf:type etype:Vaccine ;
             etype:has_vaccineName ?vaccineName ;
             etype:has_vaccineDescription ?vaccineDescription ;
             etype:has_targets ?virus .

    ## Getting virus information
    ?virus rdf:type etype:Virus ;
           etype:has_organismName ?virusName ;

    ## Extracting virusCode from its URI
    BIND(REPLACE(STR(?virus), ".*/", "") AS ?virusCode)

    ## Filtering for specific viruses
    FILTER(?virusCode = "HPV18" || ?virusCode = "HBV")
}
```

Filter query results | Compact view | Hide row numbers | Showing results from 0 to 2 of 2. Query took 0.1s, moments ago.

	virusCode	virusName	vaccineName	vaccineDescription
1	"HBV"	"Hepatitis B virus"@en	"Recombinant vaccines (Engerix-B, Recombivax HB)"	"Protect against all major Hepatitis B strains and prevent liver cancer"@en
2	"HPV18"	"Human papillomavirus"@en	"Gardasil-9 (9vHPV)"	"Protect against HPV types 16 and 18 that cause most HPV cancers."@en

Figure 15: *SPARQL query addressing CQ 4 along with the results of the query.*

CQ5 - What phenotypic traits are associated with a specific virus?

Query took: 0.1 seconds
Results obtained: 1

```
PREFIX etype: <http://knowdive.disi.unitn.it/etype#>
PREFIX rdf:   <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT DISTINCT ?virusCode ?virusName ?virusType ?trait
WHERE {
    ## Getting viruses and their associated phenotypic traits
    ?virus rdf:type etype:Virus ;
           etype:has_organismName ?virusName ;
           etype:has_hasPhenotype ?phenotype .

    ## Getting phenotype information
    ?phenotype rdf:type etype:Phenotype ;
               etype:has_virusType ?virusType ;
               etype:has_trait ?trait .

    ## Extracting virusCode from its URI
    BIND(REPLACE(STR(?virus), ".*/", "") AS ?virusCode)

    ## Filtering for a specific virus
    FILTER(?virusCode = "EBVB9")
}
```

Filter query results | Compact view | Hide row numbers | Showing results from 0 to 1 of 1. Query took 0.1s, moments ago.

	virusCode	virusName	virusType	trait
1	"EBVB9"	"Epstein-Barr virus "@en	"dsDNA"@en	"Enveloped, spherical to pleomorphic, 150-200 nm in diameter. Capsid consists of 162 capsomers and is surrounded by an amorphous tegument. Glycoproteins complexes are embedded in the lipid envelope."@en

Figure 16: *SPARQL query addressing CQ 5 along with the results of the query.*

CQ6 - What viruses target a specific host protein, and what cancers are they associated with?

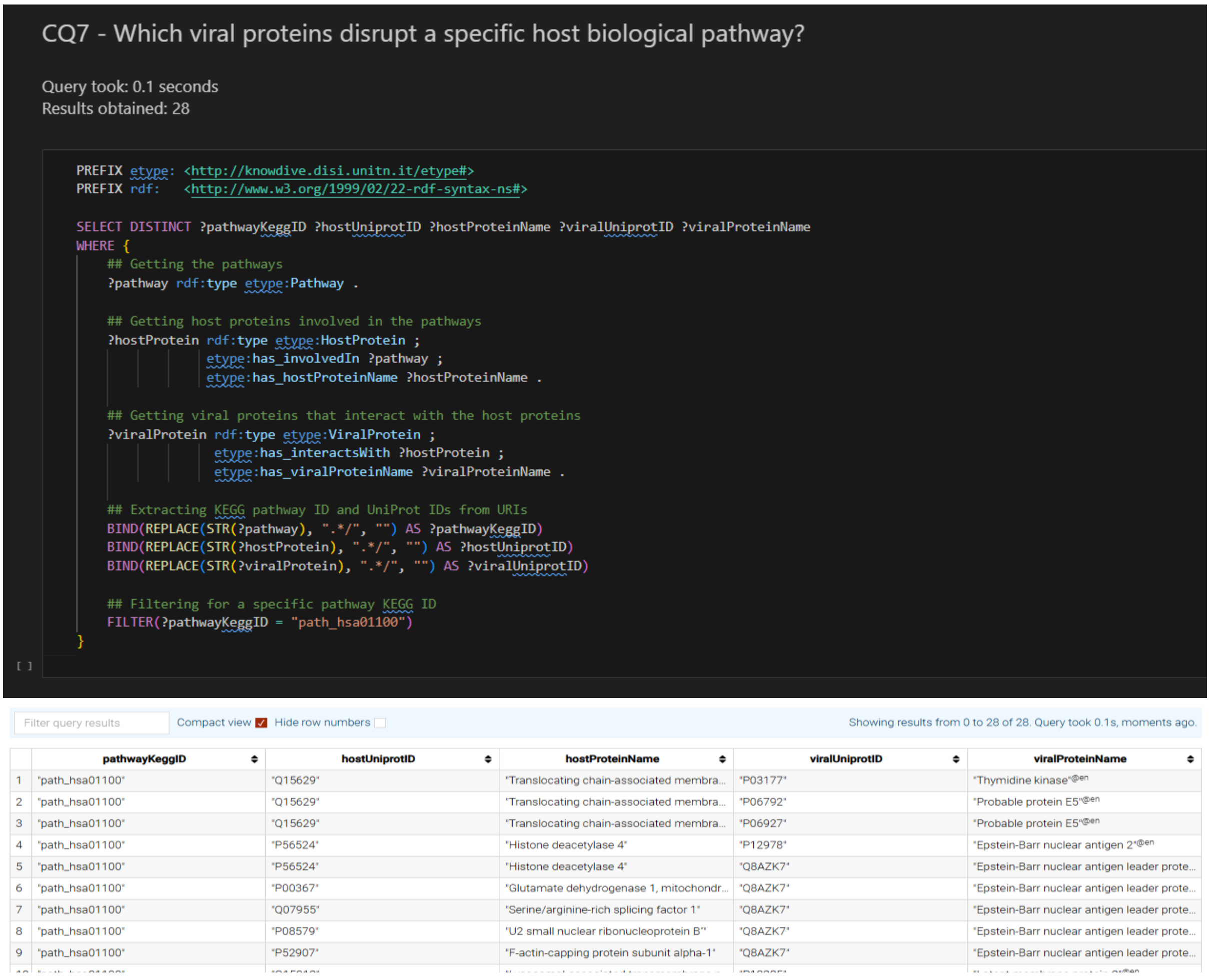
Query took: 0.1 seconds
Results obtained: 4

```
PREFIX etype: <http://knowdive.disi.unitn.it/etype#>
PREFIX rdf:   <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT DISTINCT ?hostUniprotID ?virusCode ?cancerType
WHERE {
    ## Getting viral proteins and the host proteins they interact with
    ?viralProtein rdf:type etype:ViralProtein ;
                  etype:has_interactsWith ?hostProtein .

    ## Getting the virus that encodes the viral protein and the cancer it causes
    ?virus rdf:type etype:Virus ;
           etype:has_organismName ?virusName ;
           etype:has_encodes ?viralProtein ;
           etype:has_causes ?cancer .

    ## Getting cancer type information
    ?cancer rdf:type etype:Cancer ;
            etype:has_cancerType ?cancerType .

    ## Extracting UniProt ID and virus code from URIs
    BIND(REPLACE(STR(?hostProtein), ".*/", "") AS ?hostUniprotID)
    BIND(REPLACE(STR(?virus), ".*/", "") AS ?virusCode)

    ## Filtering for a specific host protein UniProt ID
    FILTER(?hostUniprotID = "Q06330")
}
```

Filter query results Compact view Hide row numbers Showing results from 0 to 4 of 4. Query took 0.1s, moments ago.

	hostUniprotID	virusCode	cancerType
1	"Q06330"	"EBVB9"	"Burkitt lymphoma"@en
2	"Q06330"	"EBVB9"	"Hodgkin lymphoma"@en
3	"Q06330"	"EBVB9"	"Nasopharyngeal carcinoma"@en
4	"Q06330"	"EBVB9"	"Gastric carcinoma"@en

Figure 17: *SPARQL query addressing CQ 6 along with the results of the query.*



CQ7 - Which viral proteins disrupt a specific host biological pathway?

Query took: 0.1 seconds
Results obtained: 28

```
PREFIX etype: <http://knowdive.disi.unitn.it/etype#>
PREFIX rdf:   <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT DISTINCT ?pathwayKeggID ?hostUniprotID ?hostProteinName ?viralUniprotID ?viralProteinName
WHERE {
    ## Getting the pathways
    ?pathway rdf:type etype:Pathway .

    ## Getting host proteins involved in the pathways
    ?hostProtein rdf:type etype:HostProtein ;
                 etype:has_involvedIn ?pathway ;
                 etype:has_hostProteinName ?hostProteinName .

    ## Getting viral proteins that interact with the host proteins
    ?viralProtein rdf:type etype:ViralProtein ;
                  etype:has_interactsWith ?hostProtein ;
                  etype:has_viralProteinName ?viralProteinName .

    ## Extracting KEGG pathway ID and UniProt IDs from URIs
    BIND(REPLACE(STR(?pathway), ".*/", "") AS ?pathwayKeggID)
    BIND(REPLACE(STR(?hostProtein), ".*/", "") AS ?hostUniprotID)
    BIND(REPLACE(STR(?viralProtein), ".*/", "") AS ?viralUniprotID)

    ## Filtering for a specific pathway KEGG ID
    FILTER(?pathwayKeggID = "path_hsa01100")
}
```

Filter query results | Compact view | Hide row numbers | Showing results from 0 to 28 of 28. Query took 0.1s, moments ago.

	pathwayKeggID	hostUniprotID	hostProteinName	viralUniprotID	viralProteinName
1	"path_hsa01100"	"Q15629"	"Translocating chain-associated membra...	"P03177"	"Thymidine kinase"@en
2	"path_hsa01100"	"Q15629"	"Translocating chain-associated membra...	"P06792"	"Probable protein E5"@en
3	"path_hsa01100"	"Q15629"	"Translocating chain-associated membra...	"P06927"	"Probable protein E5"@en
4	"path_hsa01100"	"P56524"	"Histone deacetylase 4"	"P12978"	"Epstein-Barr nuclear antigen 2"@en
5	"path_hsa01100"	"P56524"	"Histone deacetylase 4"	"Q8AZK7"	"Epstein-Barr nuclear antigen leader prote...
6	"path_hsa01100"	"P00367"	"Glutamate dehydrogenase 1, mitochondr...	"Q8AZK7"	"Epstein-Barr nuclear antigen leader prote...
7	"path_hsa01100"	"Q07955"	"Serine/arginine-rich splicing factor 1"	"Q8AZK7"	"Epstein-Barr nuclear antigen leader prote...
8	"path_hsa01100"	"P08579"	"U2 small nuclear ribonucleoprotein B""	"Q8AZK7"	"Epstein-Barr nuclear antigen leader prote...
9	"path_hsa01100"	"P52907"	"F-actin-capping protein subunit alpha-1"	"Q8AZK7"	"Epstein-Barr nuclear antigen leader prote...

Figure 18: *SPARQL query addressing CQ 7 along with the results of the query.*

8.2 Evaluating the Purpose Definition

This evaluation examines how well the dataset aligns with the project's purpose as defined by the CQs. Coverage metrics for entity types and properties are calculated from two perspectives:

- **CQs vs Dataset coverage:** which assesses how much required knowledge is provided.
- **Dataset vs CQs coverage:** which measures the relevance of the dataset's content. Together, these metrics ensure the dataset is both sufficient and purpose focused.

8.2.1 CQs vs Dataset Coverage (ETypes)

$$cov_E^{CQ} = \frac{|ETypes_{CQs} \cap ETypes_{Dataset}|}{|ETypes_{CQs}|} = \frac{|8 \cap 8|}{|8|} = 1$$

8.2.2 CQs vs Dataset Coverage (Properties)

$$Cov_P^{CQ} = \frac{|Properties_{CQs} \cap Properties_{Dataset}|}{|Properties_{CQs}|} = \frac{|29 \cap 29|}{|29|} = 1$$

8.2.3 Dataset vs CQs Coverage (ETypes)

$$\mathrm{Cov}_E^{DS} = \frac{|\mathrm{ETypes}_{\mathrm{CQs}} \cap \mathrm{ETypes}_{\mathrm{Dataset}}|}{|\mathrm{ETypes}_{\mathrm{DSs}}|} = \frac{|8 \cap 8|}{|8|} = 1$$

8.2.4 Dataset vs CQs Coverage (Properties)

$$\mathrm{Cov}_P^{DS} = \frac{|\mathrm{Properties}_{\mathrm{CQs}} \cap \mathrm{Properties}_{\mathrm{Dataset}}|}{|\mathrm{Properties}_{\mathrm{DSs}}|} = \frac{|26 \cap 29|}{|29|} = 0.896$$

8.3 Evaluating the Language Definition

This evaluation defines how well the domain terminology of the KG aligns with the reference Language Teleontology (LTLO), which defines the standard vocabulary for the project. The assessment is based on coverage metrics that quantify the extent to which relevant terms are represented in the designed LTLO. Specifically, coverage is measured for EType terms, object property terms, and data property terms, indicating the proportion of each category that is successfully defined within the LTLO.

8.3.1 EType Term Alignment to LTLO

$$\mathrm{Cov}_{\mathrm{ET}}(\mathrm{LTLO}) = \frac{|\mathrm{ETerms}_{\mathrm{final}} \cap \mathrm{LTLO}_{\mathrm{ETerms}}|}{|\mathrm{ETerms}_{\mathrm{final}}|} = \frac{|8 \cap 8|}{|8|} = 1$$

8.3.2 Object Property Term Alignment to LTLO

$$\mathrm{Cov}_{\mathrm{OPT}}(\mathrm{LTLO}) = \frac{|\mathrm{OPTerms}_{\mathrm{final}} \cap \mathrm{LTLO}_{\mathrm{OPTerms}}|}{|\mathrm{OPTerms}_{\mathrm{final}}|} = \frac{|7 \cap 2|}{|7|} = 0.29$$

8.3.3 Data Property Term Alignment to LTLO

$$\mathrm{Cov}_{\mathrm{DPT}}(\mathrm{LTLO}) = \frac{|\mathrm{DPTerms}_{\mathrm{final}} \cap \mathrm{LTLO}_{\mathrm{DPTerms}}|}{|\mathrm{DPTerms}_{\mathrm{final}}|} = \frac{|29 \cap 12|}{|29|} = 0.41$$

8.4 Evaluating the Knowledge Definition

This evaluation examines how well the KG schema (the Teleology) aligns with the required knowledge elements, referred to as Knowledge Atoms (KATOMs). KATOMs include the essential ETypes, object properties, and data properties needed for the domain. Coverage metrics are used to assess what proportion of these required elements are represented in the designed Teleology, considering EType coverage, object property coverage, and data property coverage**.**

8.4.1 Teleontology vs KATOM (ETypes)

$$\mathrm{Cov}_E(\mathrm{KATOM}_E) = \frac{|\mathrm{KATOM}_E \cap T_E|}{|\mathrm{KATOM}_E|} = \frac{|8 \cap 8|}{|8|} = 1$$

8.4.2 Teleontology vs KATOM (Object Properties)

$$\mathrm{Cov}_{OP}(\mathrm{KATOM}_{OP}) = \frac{|\mathrm{KATOM}_{OP} \cap T_{OP}|}{|\mathrm{KATOM}_{OP}|} = \frac{|7 \cap 7|}{|7|} = 1$$

8.4.3 Teleontology vs KATOM (Data Properties)

$$\mathrm{Cov}_{DP}(\mathrm{KATOM}_{DP}) = \frac{|\mathrm{KATOM}_{DP} \cap T_{DP}|}{|\mathrm{KATOM}_{DP}|} = \frac{|29 \cap 29|}{|29|} = 1$$

8.5 Evaluating the Entity Definition

This evaluation focuses on measuring how dense and connected the KG is, both during its construction and after it is completed. Connectivity is assessed at two stages:

- On the final KG to understand its overall connectedness.
- During construction to see how much each dataset contributes to improving connectivity.

The impact of a dataset can differ between these stages because entity matching conflicts and their resolutions can change how entities are connected in the final KG.

8.5.1 Connectivity Evaluation

The connectivity of the KG can be assessed in the following ways:

- *Entity connectivity*: measures the degree to which entities are interconnected, it is calculated by using the following formula as (X, Y) is a cell matrix and OP(X) is the number of the object properties of each EType.

$$EC(X) = \frac{\sum_{Y=1}^{N}(X,Y)}{OP(X)}$$

- *Property connectivity*: measures how strongly entities are associated with their properties. It is calculated by using the following formula as (X, X) is a cell matrix and DP(X) is the number of the data properties of each EType.

$$PC(X) = \frac{(X, X)}{DP(X)}$$

	Virus	Vaccine	Phenotype	Cancer	ProteinAnnotation	ViralProtein	HostProtein	Pathway	**Sum**
Virus	36	8	19	23	20	31	0	0	101
Vaccine	8	15	0	0	0	0	3	0	11
Phenotype	19	0	56	0	0	0	8	0	27
Cancer	23	0	0	40	0	0	10	0	33
ProteinAnnotation	20	0	0	0	3470	0	696	0	716
ViralProtein	31	0	0	0	0	1210	1055	0	1086
HostProtein	0	3	8	10	696	1055	3036	593	2365
Pathway	0	0	0	0	0	0	593	8503	593
Sum	101	11	27	33	716	1086	2365	593	
									Sum
Entity Connectivity	25.25	11	27	33	716	543	788.33	593	2736.58
Properties Connectivity	9	3.75	14	13.33	1156.67	403.3	1012	118.6	2730.65

Table 5: *Representation of the connectivity matrix and the calculation of the entity and property connectivity of each EType.*

8.5.2 Sparsity Evaluation

Sparsity Quantifies how many attribute cells are zero /empty (null) in the integrated dataset. Calculated by using the following formula:

Sparsity = (0 of null values) / (total number of attribute cells)

= 14/3 = 4.667

The matrix is built using only the attributes that are shared by at least two datasets.

	virusCode	hostUniprotID	cancerID
Virus	1	0	1
Vaccine	1	0	0
Phenotype	1	0	0
Cancer	0	0	1
ProteinAnnotation	0	1	0
ViralProtein	1	1	0
HostProtein	0	1	0
Pathway	0	1	0

Table 6: *Representation of the matrix that shows the ETypes' datasets and their shared attributes*

9 Metadata Definition

All metadata for the project are stored in our GitHub repository as excel (.xlsx) files:

- project_metadata.xlsx
- language_metadata.xlsx
- knowledge_metadata.xlsx

The files contain structured definitions of all resources used and generated throughout the project lifecycle. These metadata are fully compatible with distribution through the *LiveKnowledge* category [16].

9.1 Project Metadata

This metadata encompasses information describing a digital resource, including its license, category, maintainer details, author information, associated tags, and publication date.

9.2 Language Metadata

This metadata encompasses information describing the datasets, including its version, prefix, and type; details on provenance such as the publisher, annotators, owners, validator, and generation date; licensing information; technical references including reused namespaces, reference teleontology, and UKC version; content descriptors such as keywords, data domain, and language; and project-related information including the project title and project page URL.

9.3 Knowledge Metadata

This metadata encompasses information describing datasets, including dataset keywords, publishers, creators, domains, publication timestamps, languages, versions, dataset types, ownership, access levels, licenses, and the general type of resource.

10 Open Problems and Conclusion

In this project, we built an Oncovirus KG to bring together scattered information about viruses that cause cancer and the biological mechanisms behind them. By organizing data on viruses, proteins, pathways, phenotypes, cancers, and vaccines into a single connected structure, the KG facilitates exploration of how viral infections can lead to cancer and how these processes are linked at the molecular level.

The project was carried out largely according to the initial schedule. However, the information gathering phase required more time than originally planned due to the large volume of relevant data that needed to be collected. Despite this, the main objectives of the project were achieved within the expected timeframe.

The KG was successfully implemented and used to answer the defined CQs, demonstrating its ability to support meaningful biological exploration and analysis. The results largely fulfill the initial purpose of the project and indicate that the graph is internally consistent, with all modeled entities and relationships clearly defined within its conceptual framework. Coverage by existing language ontologies was lower in some cases, which was expected given the highly specialized nature of the domain and the need for custom concepts.

Overall, this work provides a solid foundation for studying virus-induced cancers in an integrated and structured manner. In the future, the KG could be expanded to include additional viruses, virus-associated diseases, and more detailed entity attributes, thereby increasing its value for research in oncology, virology, and related scientific fields.

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